THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 8: **Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\frac{2}{5}$ of a wall is already painted. Then $\frac{2}{3}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{10}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{2}{5}$ **(d)** $\frac{11}{30}$

2. For which x is $x \div \frac{1}{3}$ less than x? 1

(a) $x > 1$ **(b)** $x < 1$

(c) no positive x **(d)** every x

3. $\frac{5}{7}$ of $2\frac{4}{5}$ is: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** $3\frac{18}{35}$

4. If $x \times \frac{2}{5} = \frac{4}{15}$, then $x \div \frac{2}{3}$ is: 1

(a) 1 **(b)** 2 **(c)** $1\frac{1}{2}$ **(d)** $\frac{1}{2}$

5. $\frac{2}{3} \div (\frac{4}{5} \div \frac{2}{5})$ equals: 1

(a) $2\frac{1}{12}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{1}{3}$

6. $\frac{4}{9}$ of a number is 20. Three times the number is: 1

(a) $67\frac{1}{2}$ **(b)** 135 **(c)** $135\frac{1}{2}$ **(d)** 270

7. A pipe fills $\frac{5}{6}$ of a tank in 20 minutes. The time to fill the whole tank (minutes) is: 1

(a) 12 **(b)** $24\frac{1}{2}$ **(c)** 48 **(d)** 24

8. $\frac{3}{8}$ of a wall is already painted. Then $\frac{3}{5}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{8}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{9}{20}$ **(d)** $\frac{1}{4}$

9. **Assertion (A):** Dividing by $\frac{2}{3}$ is the same as multiplying by $\frac{3}{2}$. 1

Reason (R): Dividing by a fraction is the same as multiplying by the fraction itself.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Multiplying a positive number by a proper fraction always gives a larger number. 1

Reason (R): $\frac{1}{2}$ of 8 is 4.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. After $\frac{3}{8}$ of a number is removed, 45 is left. Find the number and the part removed. 2

12. Find the fraction exactly halfway between $\frac{2}{5}$ and $\frac{3}{4}$, and verify that its distance from each is the same. 2

13. Simplify $\left(\frac{3}{4}+\frac{1}{4}\right) \div \left(\frac{3}{4}-\frac{1}{4}\right) \times \frac{2}{3}$. 2

14. After $\frac{4}{9}$ of a number is removed, 60 is left. Find the number and the part removed. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Ravi alone can finish a job in 12 days and Sita alone in 4 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

16. A cyclist rides at $\frac{27}{2}$ km per hour for 2 hours, then at $\frac{18}{5}$ km per hour for 5 hours. Find the total distance and the average speed for the whole journey. 3

17. A pole of 18 m is cut into pieces of $1\frac{3}{4}$ m each. How many pieces are obtained and how much is left? 3

OR

Ravi alone can finish a job in 20 days and Sita alone in 30 days. What fraction of the job do they finish together in one day, and how many days do they take together? 3

18. A cyclist rides at $\frac{45}{4}$ km per hour for 4 hours, then at $\frac{15}{2}$ km per hour for 2 hours. Find the total distance and the average speed for the whole journey. 3

OR

A pole of 30 m is cut into pieces of $3\frac{1}{2}$ m each. How many pieces are obtained and how much is left? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A table top is $\frac{3}{4}$ m long and $\frac{2}{5}$ m wide. 5

(i) Find its area.

(ii) Find its perimeter.

(iii) $\frac{1}{3}$ of the area is painted red. Find the red area.

(iv) A second table top is $1\frac{1}{2}$ m long and $\frac{2}{3}$ m wide. Find its area and how much larger it is than the first.

20. A farmer's land is 180 acres. He sells $\frac{1}{6}$ of it, gives $\frac{2}{5}$ of the remainder to his son and plants $\frac{3}{8}$ of what is left after that with sugarcane. Find each area and the area left unplanted. 5

OR

Pipe A fills a tank in 8 hours, pipe B in 24 hours, and a drain empties the full tank in 12 hours. Find the fraction filled in 1 hour with all three open, and the time to fill the empty tank. 5

END OF QUESTION PAPER — **Total: 40 Marks**